Gamma Spectroscopy of Uranium

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November 12 2024

Abstract

Gamma rays are commonly a product of nuclear reactions that happen. These reactions can either be of a decay nature or some reaction that happens due to an incoming particle impinging on the nucleus of an atom. In the case of decay, gamma rays give us information about the decaying of the parent nucleus. In this experiment gamma rays were used for a spectroscopy method such that the isotopic abundance of Uranium 235 may be found in comparison to Uranium 238. The mass that was analyzed was determined to be $0.384 \pm 0.060\%$ Uranium 235.

1 Introduction

It is well documented that there are two isotopes of the element Uranium that are stable. These being Uranium 235 (143 neutrons and 92 protons) and Uranium 238 (146 neutrons and 92 protons). The isotope Uranium 235 is of great use in nuclear reactors in that it is fissile. This property allows for the use of it's nuclear properties to add thermal energy to a water bath, which is then used to generate electricity in a nuclear power station. What is of great interest in masses of Uranium is just how much of this isotope exists in the total mass. This value is known as the enrichment amount and is the ratio of Uranium 235 to Uranium 238. Both Uranium isotopes will decay in time. When this decay transpires a different element, in an excited state, will exist. When this element's nucleus de-excites it will release a gamma ray of specific energy. This gamma ray can be detected by various means and give direct information on the parent nucleus. This in turn can be used to determine the level of enrichment when Uranium is of consideration.

2 Theory

If a mass of Uranium exists that has a certain amount of enrichment, the decays that take place over time will be of both U-235 to Th-231 and U-238 to Th-234. Naturally the produced daughter nuclei will be in an excited state which will produce gamma radiation as well as another decay to the next chain element in Uranium's decay chain. [DECAY CHAIN DIAGRAM??] Gamma rays produced both by the excited state of the daughter nucleus and the following decay can be detected and counted in the laboratory. By counting these gamma

rays and determining the energy they exist with, a direct comparison with the known results in literature can be made to determine of what origin the gamma ray was produced. For each of the decay chain elements a branching ratio may be used such that the total number of decays can be determined from the number of counted gamma rays. This alongside with the decay constant of U-235 and U-238 can be used to determine an estimate as to the ratio of one's abundance in the sample mass compared to the other.

3 Experimental Method

The experiment preformed took place at the Edwards Accelerator Laboratory in Athens, Ohio. A mass of Uranium was used for the process of gamma ray spectroscopy. In order to determine in later calculations the level of enrichment of the mass, a Germanium detector was utilized to detect gamma rays emitted from the mass. The Germanium detectors are known for their high level of resolution, making detected gamma ray energies relatively easy to distinguish between with low error. The experimental setup consisted of one Germanium detector paired with an ADC which would then feed data to a PiDAQ data acquisition system where the data would be appropriately binned. Adjustments were made on the ADC gain to set what was believed to be the best gain setting for corresponding peaks showing decays of U-235 and U-238.

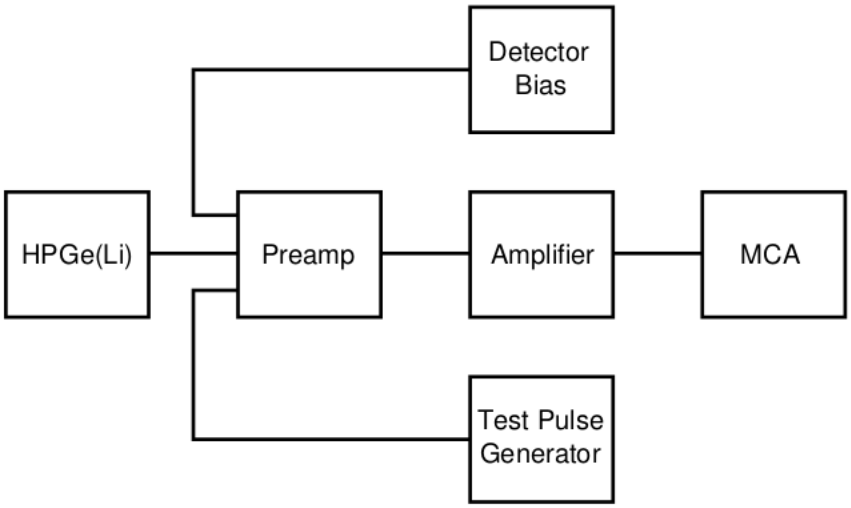


Figure 1: Block Diagram of the Experimental Electronics

With this value now set, and to no longer be adjusted for the rest of the experiment, a linear energy calibration was determined with a set of three calibration sources. The sources used were Na-22, Cs-137, and Co-60. These sources have been documented in the literature and gave peaks ranging from around 551keV up to 1.3MeV.

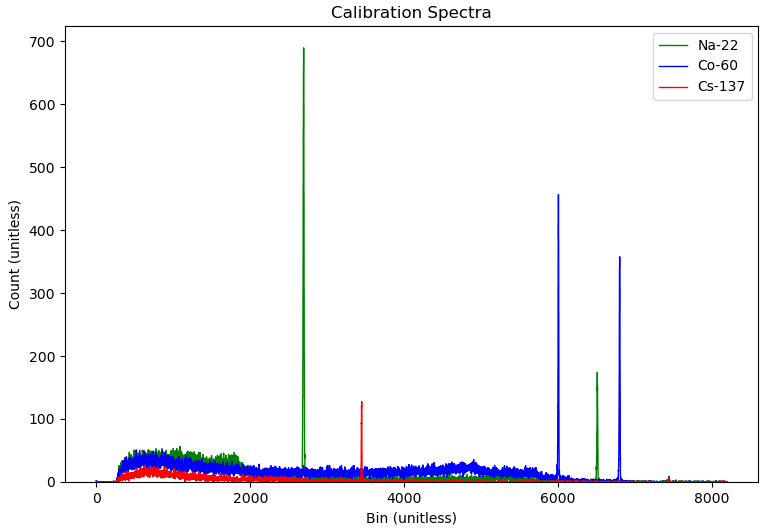


Figure 2: Spectra Recorded by Germanium Detector from Calibration Sources

Each peak from the spectra shown in Figure 2 was analyzed such that its corresponding centroid and error of the centroid would be used to determine a linear energy to bin calibration relation. The points described by the centroids of the peaks and the known energies from the literature were used to fit a linear regression line. The resulting slope and constant energy shift is tabulated below.

$$E(x) = mx + b$$

Parameter	Value	Error	Unit
m	0.201539	0.000001	keV/bin
b	-30.7427	0.0066	keV

The energy calibration's linear nature will be used to describe the energy of the gamma rays detected with the Germanium detector. The next factor that was taken into consideration was not only does the energy of the peaks in the Uranium gamma spectrum give insight into what isotope is decaying, but the Germanium detector does not accurately record the full range of events that transpired. This is due to the detector's efficiency not being a constant value. In order to describe this phenomenon by means of determining an efficiency versus bin, and thus energy, function, an Eu-152 source was used.

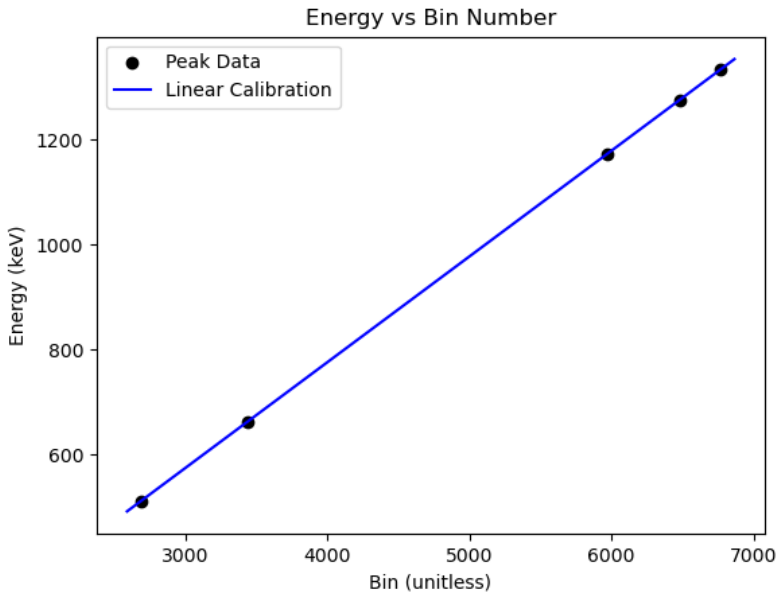


Figure 3: Linear Energy Calibration Fit

The Europium source was placed in front of the detector in the lab for a period of time that allowed for the clear identification of peaks to be seen. As with the earlier calibration sources, the same statistics were preformed on each peak that was identified with the known literature energy for the Europium spectra. By utilizing the counts in each peak, as well as deciding on an initial peak to define an efficiency as 100%, an efficiency curve that had an exponential decay nature was fit to the data. However, once this was done it was noticed that a slight deviation from the fit occurred for peaks at lower bin number rather than peaks at higher bin number in the spectra. Upon reconsideration of the original Eu-152 spectra it was considered that this deviation may be due to the spectra's background counts for lower bin number, and thus the lower energy region.

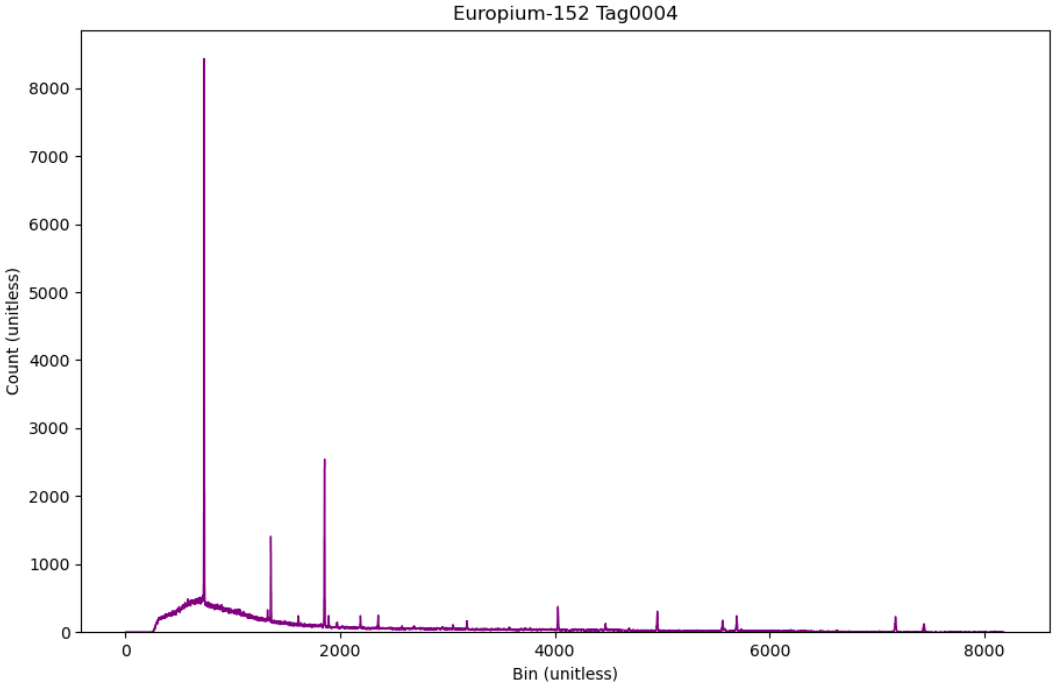


Figure 4: Spectra from Eu-152 Sources

Due to this fact, an algorithm was constructed in the Python programming language to remove the background counts from the spectra's peaks, and thus adjust the statistics for a more accurate account of what the true efficiency was. This was done by implementing a Python package to fit the spectra data to a linear function plus a Gaussian. In this manor the liner portion could then be removed from the original peak data and the statistics run again. Once this was done for all of the peaks in the Eu-152 spectrum pertaining to the efficiency calculation, a exponential decay fit was preformed again. This lead to the detector efficiency relation with energy that was desired. The results of this fit are tabulated below.

$$f(x) = Ae^{-Bx-C}$$

Parameter	Value	Error	Unit
A	1.76511	0.09788	none
B	0.00091	0.00006	1/bin
C	-0.10106	0.01389	none

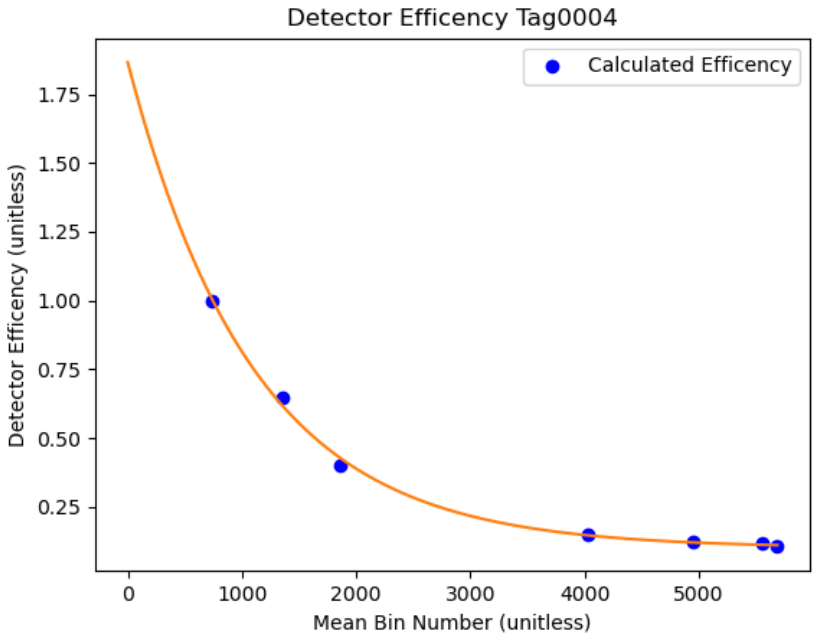


Figure 5: Efficiency Exponential Fit With Background Removal

Considering now that the efficiency values can be determined for any peak's centroid, the expected number of gamma ray events can be calculated by simply taking into account the efficiency function's value at that energy.

Having the total number of expected events on the detector now allowed us to use the branching ratios found in the literature for each defined peak. This paired with the decay half-lives for U-238 and U-235 would then allow us to estimate the ratio of U-235 isotopes to the U-238 isotopes.

4 Results and Analysis

Each number of expected events the detector recorded paired with the literature's branching ratio values leads to the number of total decays that occurred. However simply taking the ratio of these values would not be the resulting enrichment amount. Instead what must be considered is this ratio multiplied by the quotient factor of each isotopes radioactive decay constants. In this experiment the half lives were used instead of the decay constants, however the result is the same due to the fact many of the constants cancel in the ratio since the resulting quotient is unit-less.

Parent Isotope	Energy (keV)	Intensity (%)
U-235	143.76	10.96
U-235	163.33	5.08
U-235	185.715	57.2
U-235	202.11	1.08
U-238	742.81	2.06
U-238	766.36	0.294
U-238	786.27	0.0486
U-238	1001.7	0.838

Peak	Energy (keV)	Error (keV)	Efficiency (%)	Background	Exp. Decays
1	140.19058	0.00813	91.924	20551	132071
2	160.24895	0.01095	84.870	14447	184603
3	183.07183	0.00702	77.691	15829	204236
4	203.04374	0.01098	71.879	6765	213640
5	745.96844	0.02628	15.493	1499	291074
6	769.63295	0.01615	14.955	1720	9352601
7	789.65121	0.04391	14.535	887	12330047
8	1006.1718	0.01360	11.783	557	8136172

Using the average of each expected decay count multiplied with its corresponding branching ratio for both U-238 and U-235, a ratio of the two was taken. This value was then adjusted with the quotient of the U-235 half life to the U-238 half life resulting in a value of $0.384 \pm 0.060\%$. This number obviously shows a very low level of enrichment, which was to be expected of the sample mass in the laboratory. It is believed that the resulting error mainly stems from the energy to bin calibration. While it's true that there was quite little error in the resulting function, it would be to the benefit of the experiment if the calibration data were to be recorded for a longer period of time. This can especially be seen in the Cs-137 peak as it has very low counts compared to the rest of the calibration sources' gamma ray peaks. It should also be taken into consideration that a different form of background subtraction could have been used in the Python script. Instead of a linear function describing the background counts, a polynomial fit may have been more appropriate.

5 Conclusion

In conclusion the spectra generated via gamma rays from the exponential decay of nuclei can give great insight into the properties of a mass that is built of atoms containing that nuclei. As seen for the case in this experiment, pertaining to Uranium, the level of enrichment can be directly determined by this gamma ray spectroscopy. Naturally the statistics of each peak will determine heavily on the characteristics of the detector used, such as resolution, efficiency, and various other properties specific to differing detectors, yet this can all be accounted for to get key insight to the material of interest.

6 References

1 G.Jensen P.M. King, and J. Roche, *PHYS6751: Nuclear Graduate Lab Lab manual v24.5,* (Ohio University, 2024)

2 *"National Nuclear Data Center", NNDC* https://www.nndc.bnl.gov